

Open Mathematical Olympiad for University Students (OMOUS-2021)

Individual Olympiad

Group A

Problem 1 Let a_n number of perfect squares of between n^3 and $(n+1)^3$.

a) Prove that $a_n \geq 1$ for every $n \in \mathbb{N}$

b) Calculate $\lim_{n \rightarrow \infty} \frac{a_n}{\sqrt{n}}$

Leonid Genze (Tomsk State University, Russian Federation)

Solution.

a) Number of perfect squares, not exceeding m is equal to $[\sqrt{m}]$, as

$$1^2, 2^2, \dots, [\sqrt{m}]^2 \leq \sqrt{m}^2 = m, \text{ but } ([\sqrt{m}] + 1)^2 > \sqrt{m}^2 = m.$$

In this way, number of perfect squares a_n between n^3 and $(n+1)^3$ equals to

$$[\sqrt{(n+1)^3}] - [\sqrt{n^3}].$$

$$b_n = \sqrt{(n+1)^3} - \sqrt{n^3} = \frac{(n+1)^3 - n^3}{\sqrt{(n+1)^3} + \sqrt{n^3}} = \frac{3n^2 + 3n + 1}{\sqrt{(n+1)^3} + \sqrt{n^3}}$$

$$b_n > \frac{3n^2}{2\sqrt{n^3} + \sqrt{n^3}} = \frac{3n^2}{3\sqrt{n^3}} = \sqrt{n^3} > 2 \text{ for } n \geq 2.$$

$$a_n = [\sqrt{(n+1)^3}] - [\sqrt{n^3}] \geq \sqrt{(n+1)^3} - 1 - \sqrt{n^3} = b_n - 1 > 2 - 1 = 1$$

For $n = 1$, $a_n = 1$. Then, there is at least one perfect square between n^3 and $(n+1)^3$.

$$a_n \leq \sqrt{(n+1)^3} - \sqrt{n^3} + 1 = b_n + 1$$

$$\text{b) } \lim_{n \rightarrow \infty} \frac{a_n}{\sqrt{n}} = \lim_{n \rightarrow \infty} \frac{b_n}{\sqrt{n}} = \lim_{n \rightarrow \infty} \frac{3n^2 + 3n + 1}{\sqrt{n}(\sqrt{(n+1)^3} + \sqrt{n^3})} = \frac{3}{2}$$

Problem 2

For $A, B, C \in M_n(\mathbb{C})$ prove that the following statements are equivalent:

$$\text{a) } (A + I)(B - I) = (B + I)(C - I) = (C + I)(A - I) = O_n$$

$$\text{b) } A^2 = B^2 = C^2 = I \text{ and } \text{Inv}A = \text{Inv}B = \text{Inv}C$$

Where for $M \in M_n(\mathbb{C})$, we denote $\text{Inv}M = \{X \in M_{n,1}(\mathbb{C}) : M \cdot X = -X\}$

Vasile Pop (Technical University of Cluj-Napoca, Romania),

Mihai Opincariu (Romania)

Solution.

For $M \in M_n(\mathbb{C})$, denote $M_1 = M + I_n$. With this notation, (1) is equivalent to

$$A_1 B_1 = 2A_1 \quad B_1 C_1 = 2C_1 \quad C_1 A_1 = 2C_1 \quad (1')$$

And (2) is equivalent to

$$A_1^2 = 2A_1, \quad B_1^2 = 2C_1, \quad C_1^2 = 2C_1, \quad \text{Ker}A_1 = \text{Ker}B_1 = \text{Ker}C_1, \quad (2')$$

Since

$InvM = \{X \in M_{n,1}(C): M \cdot X = -X\} = \{X \in M_{n,1}(C): M_1 \cdot X = 0\} = KerM_1$.
We need to prove that (1') $\Leftrightarrow$ (2').

(1') $\Rightarrow$ (2')

$$\left. \begin{array}{l} A_1 B_1 = 2A_1 \Rightarrow A_1 B_1 C_1 = 2A_1 C_1 \\ B_1 C_1 = 2B_1 \Rightarrow A_1 B_1 C_1 = 2A_1 B_1 = 4A_1 \end{array} \right\} \Rightarrow A_1 C_1 = 2A_1 \quad (1_a)$$

$$\left. \begin{array}{l} C_1 A_1 = 2C_1 \Rightarrow A_1 C_1 A_1 = 2A_1 C_1 = 4A_1 \\ (1_a) \Rightarrow A_1 C_1 A_1 = 2A_1^2 \end{array} \right\} A_1^2 = 2A_1$$

By the symmetry of the hypothesis with respect to A, B and C , it follows that also $B_1^2 = 2B_1$ and $C_1^2 = 2C_1$. Also,

$$\begin{aligned} X \in KerB_1 &\Leftrightarrow B_1 \cdot X = 0 \Rightarrow \left. \begin{array}{l} A_1 B_1 \cdot X = 0 \\ A_1 B_1 = 2A_1 \end{array} \right\} \Rightarrow 2A_1 \cdot X_1 = 0 \Rightarrow A_1 \cdot X_1 = 0 \Rightarrow \\ &\Rightarrow X \in KerA_1. \end{aligned}$$

So $KerB_1 \subseteq KerA_1$. By the symmetry of the hypothesis, it follows that $KerC_1 \subseteq KerB_1$ and $KerA_1 \subseteq KerC_1$, hence

$$KerA_1 \subseteq KerC_1 \subseteq KerB_1 \subseteq KerA_1,$$

meaning that

$$KerA_1 = KerB_1 = KerC_1$$

And (2') is completely proved.

$$\begin{aligned} (2') \Rightarrow (1') \quad A_1^2 = 2A_1 &\Leftrightarrow A_1(A_1 - 2I_n) = O_n \Leftrightarrow A_1(A_1 - 2I_n) \cdot X = 0, \text{ for all } X \in M_{n,1}(C) \\ &\Leftrightarrow (A_1 - 2I_n) \cdot X \in KerA_1, \text{ for all } X \in M_{n,1}(C) \Leftrightarrow (A_1 - 2I_n) \cdot X \in KerC_1, \\ &\text{for all } X \in M_{n,1}(C) \Leftrightarrow C_1(A_1 - 2I_n) \cdot X = 0, \text{ for all } X \in M_{n,1}(C) \Leftrightarrow C_1(A_1 - 2I_n) = \\ &O_n \Leftrightarrow C_1 A_1 = 2C_1. \end{aligned}$$

By the symmetry of the hypothesis, the other two relations are also satisfied.

Problem 3

If is the $\{F_n\}$ Fibanacci sequence, then evaluate the sum:

$$\sum_{n=1}^{\infty} (-1)^n \frac{F_{F_{n-1}}}{F_{F_n} F_{F_{n+1}}}$$

Khakimboy Egamberganov (UPMC, Sorbonne University, France)

Solution.

$$\text{Answer: } \frac{1-\sqrt{5}}{2}$$

We have the following beautiful property of the Fibonacci sequence

$$F_n F_{m+k} - F_{n+k} F_m = (-1)^m F_k F_{n-m}$$

which is called the subtraction formula. Actually, for the Fibonacci sequence we know its n^{th} term formula

$$F_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$$

And we can define F_{-n} as $-F_n$ for $n > 0$.

By the subtraction formula, we can find

$$(-1)^{F_n} F_1 F_{F_{n+1}-F_n} = F_{F_{n+1}} F_{F_n+1} - F_{F_{n+1}+1} F_{F_n}$$

and

$$(-1)^{F_n} \frac{F_{F_{n-1}}}{F_{F_n} F_{F_{n+1}}} = \frac{F_{F_{n+1}}}{F_{F_n}} - \frac{F_{F_{n+1}+1}}{F_{F_{n+1}}}$$

So

$$\sum_{n=1}^{\infty} (-1)^{F_n} \frac{F_{F_{n-1}}}{F_{F_n} F_{F_{n+1}}} = 1 - \lim_{n \rightarrow \infty} \frac{F_{F_{n+1}}}{F_{F_n}} = 1 - \frac{1+\sqrt{5}}{2} = \frac{1-\sqrt{5}}{2}.$$

Here, we used n^{th} for Fibonacci sequence which implies $F_n \sim \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n$ as $n \rightarrow \infty$.

Problem 4

Let $\xi_1, \xi_2, \xi_3, \dots, \xi_n, \dots$ be sequence of identically distributed continuously randomly variables and $v = \min\{k: \xi_{k-1} > \xi_k\}$. Find mathematical expectation $E(v)$.

Orazmammet Annaorazov (Magtymguly Turkmen State University, Turkmenistan)

Solution. Since the random variables $\xi_1, \xi_2, \xi_3, \dots, \xi_n, \dots$ are continuous, therefore $P\{\xi_i = \xi_j\} = 0$ for $i \neq j$. Let's write the law of distribution of a discrete random variable v

$$\begin{array}{ccccccc} v & 2 & 3 & & n \\ P & P_2 & P_3 & \cdots & P_n \end{array}$$

Where $P_n = P\{v = n\} = P\{\xi_1 \leq \xi_2 \leq \dots \leq \xi_{n-1} > \xi_n\}$.

Then,

$$\begin{aligned} P_n &= P\{\xi_1 \leq \xi_2 \leq \dots \leq \xi_{n-1} \leq \xi_n\} + P\{\xi_1 \leq \xi_2 \leq \dots \leq \xi_{n-1} > \xi_n\} - \\ &- P\{\xi_1 \leq \xi_2 \leq \dots \leq \xi_{n-1} \leq \xi_n\} = P\{\xi_1 \leq \xi_2 \leq \dots \leq \xi_{n-2} \leq \xi_{n-1}\} - \\ &- P\{\xi_1 \leq \xi_2 \leq \dots \leq \xi_{n-1} \leq \xi_n\} = \frac{1}{(n-1)!} - \frac{1}{n!} = \frac{n-1}{n!}. \end{aligned}$$

So,

$$Mv = \sum_{n=2}^{\infty} n P_n = \sum_{n=2}^{\infty} n \frac{n-1}{n!} = \sum_{n=2}^{\infty} \frac{1}{(n-2)!} = e$$

Problem 5

Let $A(a_{ij}) \in M_n(\mathbb{R})$ symmetric matrix such that $a_{ii} = 0$. If at least $\left\lceil \frac{n+n\sqrt{4n-3}}{2} \right\rceil$ entries of matrix A equal to 1 and other elements equal to 0, then prove that matrix A has at least

one $\begin{pmatrix} 1 & 1 & a & 1 \\ 1 & 1 & 1 & b \\ c & 1 & 1 & 1 \\ 1 & d & 1 & 1 \end{pmatrix}$ minor. Where $a, b, c, d \in \{0; 1\}$.

Murat Chashemov (Turkmenistan)

Solution. Let G be n vertex planar graph with vertices $d_1, d_2, d_3, \dots, d_n$ that has only vertices without connected any connection i.e edges. Let's correspond a graph for any such matrix. If matrix A has elements $a_{ij} = a_{ji} = 1$ then we connect vertices d_i and d_j of graph G , elsewhere there is no edge between vertices d_i and d_j if $a_{ij} = a_{ji} = 0$ for the diagonal elements this graph also satisfies our correspondence $a_{ii} = 0$ because we can't connect vertex d_i with himself. So we get new planar graph G_A for any such matrix. It's easy to see that there is bijection with such matrices and graphs. If we translate our problem to the graphs we need to prove that:

For every graph G_A with vertices $d_1, d_2, d_3, \dots, d_n$ ($n \geq 4$) and with edges more than $\left\lceil \frac{n+n\sqrt{4n-3}}{4} \right\rceil$ there is a 4 cycle.

If we prove above we get some chain $d_{i_1}, d_{i_2}, d_{i_3}, d_{i_4}, d_{i_1}$ and this chain corresponds to the

some minor of form $\begin{pmatrix} 0 & 1 & * & 1 \\ 1 & 0 & 1 & * \\ * & 1 & 0 & 1 \\ 1 & * & 1 & 0 \end{pmatrix}$ that located in the main diagonal of matrix A . So for

matrix $A + I$ we can find minor of form $\begin{pmatrix} 1 & 1 & * & 1 \\ 1 & 1 & 1 & * \\ * & 1 & 1 & 1 \\ 1 & * & 1 & 1 \end{pmatrix}$.

Let's to prove that our graph contains 4 cycle. Let us count, in two different ways, the number of triples (c, a, b) where a, b, c are vertices such that c is connected to both a and b . For a fixed vertex c , there are $d(c)^2 - d(c)$ possibilities for the pair (a, b) , where $d(c)$ denotes the degree of vertex c . It follows that there are at least $\sum_c (d(c)^2 - d(c))$ triples. By the Cauchy-Schwarz inequality, if $m > \left\lceil \frac{n+n\sqrt{4n-3}}{4} \right\rceil$ represents the number of edges of the graph, then:

$$\begin{aligned}\sum_c d(c)^2 - d(c) &\geq \sum_c d(c)^2 - \sum_c d(c) > \sum_c d(c)^2 - m \geq \frac{(\sum_c d(c))^2}{n} - 2m \\ &> \frac{4m^2}{n} - 2m\end{aligned}$$

Now, if there are no 4-cycles, then for fixed vertices a and b there is at most one vertex c that appears in a triple (a, b, c) . Hence the number of such triples is equal to the number of pairs so we obtain at most $n(n-1)$ triples. It follows that $\frac{4m^2}{n} - 2m < n^2 - n$, which implies that $m < \left\lceil \frac{n+n\sqrt{4n-3}}{4} \right\rceil$ contradiction. Hence there must be some 4 cycle.

Problem 6

Let f be a nonnegative and decreasing function on $[0, \infty)$ and let g be a function defined on $[0, \infty)$ such that $0 \leq g(x) \leq 2021$ with $\int_0^\infty g(x)dx = 2021$. Then prove that

$$\int_0^\infty f(x)g(x)dx \leq 2021 \int_0^1 f(x)dx$$

Pirmyrat Gurbanov (International University for the Humanities and Development, Turkmenistan)

Solution. Problem is particular case of Steffensen's Inequality.

Steffensen's Inequality: Let f be a nonnegative and decreasing function on $[0, \infty)$ and let g be a function defined on $[0, \infty)$ such that $0 \leq g(x) \leq A$ ($A > 0$) with $\int_0^\infty g(x)dx < \infty$. Then

$$\int_0^\infty f(x)g(x)dx \leq A \int_0^\lambda f(x)dx$$

where

$$\lambda = \frac{1}{A} \int_0^\infty g(x)dx$$

Proof: We have

$$\begin{aligned}&\int_0^\infty f(x)g(x)dx - A \int_0^\lambda f(x)dx \\ &= - \int_0^\lambda (A - g(x))(f(x) - f(\lambda))dx - \int_\lambda^\infty g(x)(f(\lambda) - f(x))dx \leq 0\end{aligned}$$

Individual Olympiad

Group B

Problem 1

The first trade market was opened in the Nusai fortress city. There are three goods: a jug (J), a dagger (D) and an arrow (A), which could be sold at the market, and there are coins (C). Anyone who comes to the market can make an exchange in the following order:

$$1 J=2D, 1A=6J, 11D =1A, 1C=10D \text{ and } 15D=1C$$

And the exchange $C \leftrightarrow J$ и $C \leftrightarrow A$ are prohibited.

Mitridat came to the market with 100 coins. Can he leave the market with 2021 coins?

Guychgeldi Kutlyyev (Turkmen State Institute of Finance, Turkmenistan)

Solution:

Let us consider the repetition $D \rightarrow A \rightarrow J \rightarrow D$. Here, if there are initially 11 daggers, then we get:

$11D = 1A = 6J = 12D$, having 12 daggers from 11 daggers. Buying 1000 daggers for 100 C, we can exchange 90 times 11 daggers as in the scheme above. This way, we get 90 extra daggers at the first stage (1090 daggers in total). Now, breaking down the 1090 daggers into 11-dagger groups, then, we get 99 extra daggers at the next stage (now 1189 daggers in total), and so on. Therefore, if the number of the initial 1000 daggers reach 30315, they can be exchanged for 2021 C.

If we use the linear growth algorithm of our capital (daggers), the problem can be solved completely.

Problem 2

Find all possible values a and b such that

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}^4 = \begin{pmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix}$$

Pirmyrat Gurbanov (International University for the Humanities and Development, Turkmenistan)

Shahlo Hasanova (Leuphana University of Lüneburg, Germany)

Solution:

Let $a^2 - b^2 = x$ and $2ab = y$, then

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}^2 = \begin{pmatrix} a^2 - b^2 & -2ab \\ 2ab & a^2 - b^2 \end{pmatrix} = \begin{pmatrix} x & -y \\ y & x \end{pmatrix}$$

and

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}^4 = \begin{pmatrix} x & -y \\ y & x \end{pmatrix}^2 = \begin{pmatrix} x^2 - y^2 & -2xy \\ 2xy & x^2 - y^2 \end{pmatrix} = \begin{pmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix}$$

$$\begin{cases} x^2 - y^2 = 1 \\ 2xy = \sqrt{3} \end{cases}$$

Solving system of equation, we get four different solution

$$\begin{cases} x = \frac{\sqrt{3}}{\sqrt{2}} \\ y = \frac{1}{\sqrt{2}} \end{cases} \quad \begin{cases} x = -\frac{\sqrt{3}}{\sqrt{2}} \\ y = -\frac{1}{\sqrt{2}} \end{cases} \quad \begin{cases} x = \frac{i}{\sqrt{2}} \\ y = -\frac{\sqrt{3}i}{\sqrt{2}} \end{cases} \quad \begin{cases} x = -\frac{i}{\sqrt{2}} \\ y = \frac{\sqrt{3}i}{\sqrt{2}} \end{cases}$$

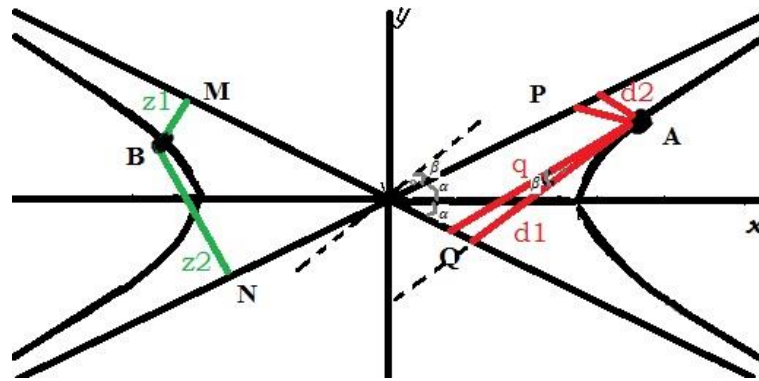
By using same idea, we can easy get 16 different solutions of equation.

Problem 3

Let's choose two arbitrary points A and B from the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Draw two straight lines through the point A parallel to the asymptotes, which intersect these asymptotes at points P and Q. Through the point B let's restore two perpendiculars to the asymptotes and denote intersection points by M and N. Compute upper bound of the ratio of the product $BN \cdot BM$ to the area of the parallelogram APOQ, if O is an origin.

Pirmyrat Gurbanov (International University for the Humanities and Development, Turkmenistan)

Solution:



Product of distances from any point of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ up to two asymptotes, is a constant value, equal to

$$\frac{a^2 b^2}{a^2 + b^2} \quad (1)$$

Let's prove it. The equation of the asymptotes of hyperbola is:

$$y = \pm \frac{b}{a} x.$$

According to the equation of normal, for first asymptote, distance (z_1) is equal to:

$$z_1 = \frac{a}{\sqrt{a^2+b^2}} \left(y + \frac{b}{a}x \right)$$

For second asymptote, distance (z_2) is equal to

$$z_2 = \frac{a}{\sqrt{a^2+b^2}} \left(y - \frac{b}{a}x \right)$$

$$\begin{aligned} z_1 z_2 &= \frac{a}{\sqrt{a^2+b^2}} \left(y + \frac{b}{a}x \right) \frac{a}{\sqrt{a^2+b^2}} \left(y - \frac{b}{a}x \right) = \frac{a^2}{a^2+b^2} \left(y^2 - \frac{b^2}{a^2}x^2 \right) \\ &= \frac{a^2 b^2}{a^2+b^2} \left(\frac{y^2}{b^2} - \frac{x^2}{a^2} \right) = \frac{a^2 b^2}{a^2+b^2} \end{aligned}$$

Area of parallelogram, bounded by the asymptotes of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and straight lines drawn through any point parallel to the asymptotes, is a constant value, equal:

$$S = \frac{ab}{2} \quad (2)$$

Now, let's prove it. It's known that area of parallelogram is $S = q * h$, where q is base, and h is height of parallelogram. Height of the parallelogram is equal to d_2 , base of parallelogram equal to $q = \frac{d_1}{\cos \beta}$, where β is defined as:

$$\begin{aligned} \tan \alpha &= \frac{b}{a}, \quad \frac{\pi}{2} = 2\alpha + \beta, \quad \beta = \frac{\pi}{2} - 2 \tan^{-1} \frac{b}{a} \\ \cos \beta &= \cos \left(\frac{\pi}{2} - 2 \tan^{-1} \frac{b}{a} \right) = \sin \left(2 \tan^{-1} \frac{b}{a} \right) = \frac{2ab}{a^2+b^2} \end{aligned}$$

Now we have the area of parallelogram:

$$S = q * h = \frac{d_1}{\cos \beta} * d_2 = \frac{d_1 d_2}{\frac{2ab}{a^2+b^2}}$$

Here, d_1 and d_2 is the distance from point of hyperbola to the asymptote, then their product equals to (as shown in (1)):

$$d_1 d_2 = \frac{a^2 b^2}{a^2+b^2}.$$

Then we have the area of parallelogram as:

$$S = \frac{\frac{a^2 b^2}{a^2+b^2}}{\frac{2ab}{a^2+b^2}} = \frac{ab}{2}$$

The question was to compute upper bound of $BN \cdot BM$ to the area of the parallelogram APOQ, then we have:

$$\frac{BN \cdot BM}{S_{APOQ}} = \frac{\frac{a^2 b^2}{a^2+b^2}}{\frac{ab}{2}} = \frac{2ab}{a^2+b^2} \leq 1.$$

Problem 4

Define the sequence $\{a_n\}$ by $a_0 = 1, a_1 = 1$ and $a_n = a_{n-1} + 2(n-1)a_{n-2}$
Find sum of the series

$$\sum_{n=0}^{\infty} \frac{a_n}{n!}$$

Vasile Pop, Mircea Rus (Technical University of Cluj-Napoca, Romania)

Solution:

Consider the (formal) power series $S(x) = \sum_{n=0}^{\infty} \frac{a_n x^n}{n!}$. Since our numerical series is $S(1)$, we need to make sure that S has a radius of convergence large enough for $S(1)$ to make sense. We can easily prove by induction that

$$0 \leq b_n := \frac{a_n}{n!} \leq \frac{1}{2^{n-1}} \text{ for all } n \geq 0.$$

The positivity is obvious. Also,

$$b_0 = a_0 = 1 \leq 2 = \frac{1}{2^{-1}}, b_1 = a_1 = 1 \leq \frac{1}{2^{1-1}},$$

While the inductive step is as follows:

$$\begin{aligned} b_n = \frac{a_n}{n!} &= \frac{a_{n-1} + 2(n-1)a_{n-2}}{n!} = \frac{1}{n}(b_{n-1} + 2b_{n-2}) \leq b_{n-1} + 2b_{n-2} \leq \frac{1}{2^{n-2}} + \frac{1}{2^{n-3}} \\ &= \frac{1}{2^{n-1}} \end{aligned}$$

The relation (1) is enough to ensure that the radius of convergence for S is at least 2.

Next,

$$\begin{aligned} S'(x) &= a_1 + \sum_{n=2}^{\infty} \frac{a_n x^{n-1}}{(n-1)!} = a_1 + \sum_{n=2}^{\infty} \frac{a_{n-1}}{(n-1)!} x^{n-1} + 2x \sum_{n=2}^{\infty} \frac{a_{n-2}}{(n-2)!} x^{n-2} = a_0 + \\ &\sum_{m=1}^{\infty} \frac{a_m}{m!} x^m + 2x \sum_{m=0}^{\infty} \frac{a_m}{m!} x^m = S(x) + 2xS(x) = (1+2x)S(x) \end{aligned}$$

So S satisfies the differential equation:

$$S'(x) - (1+2x)S(x) = 0$$

The integrating factor is $e^{-(x+x^2)}$, which leads to the solution

$$S(x) = Ce^{x+x^2}, C \in \mathbb{R} \text{ constant}$$

Since $S(0) = a_0 = 1$, it follows that $C = 1$. Finally, letting $x = 1$, we obtain

$$\sum_{n=0}^{\infty} \frac{a_n}{n!} = S(1) = e^2$$

Problem 5

Let $A_0 \neq aI_2$ and $A_0, A_1, \dots, A_n \in M_2(\mathbb{R})$ matrices such that $A_0 A_k = A_k A_0$
 $\forall k = 1, 2, \dots, n$. Then prove that $\det(\sum_{k=1}^n A_k^2) \geq 0$.

Yusupbay Ermetov (Turkmen Agricultural Institute, Turkmenistan)

Solution:

We can easily get $A_0 A_k = A_k A_0$ $k = 1, 2, \dots, n$, from the equation $A_k = a_k A_0 + b_k I_2$, $k = 1, 2, \dots, n$.

If all $a_k = 0$, then $\sum_{k=1}^n A_k^2 = (\sum_{k=1}^n b_k^2) \cdot I_2$. So $\det(\sum_{k=1}^n A_k^2) = (\sum_{k=1}^n b_k^2)^2 \geq 0$

If $a = \sum_{k=1}^n a_k^2 \neq 0$, then

$$\sum_{k=1}^n A_k^2 = \left(\sum_{k=1}^n a_k^2 \right) A_0^2 + 2 \left(\sum_{k=1}^n a_k b_k \right) A_0 + \left(\sum_{k=1}^n b_k^2 \right) I_2 = P(A_0)$$

Here $P(X) = (\sum_{k=1}^n a_k^2) X_0^2 + 2(\sum_{k=1}^n a_k b_k) X_0 + (\sum_{k=1}^n b_k^2)$ polynomial equation. Its discriminant

$$\Delta = 4 \left(\sum_{k=1}^n a_k b_k \right)^2 - 4 \left(\sum_{k=1}^n a_k^2 \right) \cdot \left(\sum_{k=1}^n b_k^2 \right)$$

From Cauchy inequality equation is non-positive. Polynomial we can write form

$$P(z) = \alpha \cdot (z - z_0)(z - \bar{z}_0)$$

Where $z_0 \in \mathbb{C}$ and $\alpha \in \mathbb{R}$, Then

$$P(A_0) = \alpha \cdot (A_0 - z_0 E_2)(A_0 - \bar{z}_0 E_2)$$

Where $\det P(A_0) = \alpha^2 |\det(A_0 - z_0 I_2)|^2 \geq 0$

Problem 6

Aman and Serdar play a game. Aman places 140 ones and 140 zeros on two concentric circles at equal distances so that there are 140 numbers in each. The numbers of the inner circle and the numbers of the outer circle lie in the rays passing through the center, and these numbers are called "corresponding numbers". If the corresponding numbers are different, we call these numbers a "good pair". If Serdar can rotate the inner circle so that the number of good pairs is not less than 70, he will win the game (rotate it so that the corresponding numbers are formed). If Serdar can't do this process so, Aman will win the game. Which of these players will win?

Palvan Agamyradov (Magtymguly Turkmen State University, Turkmenistan)

Solution 1 (Author): Serdar will win the game. May Aman have placed these numbers in any form. Let us prove that Serdar can rotate small circle so that the number of good pairs is not less than 70. Serdar replaces all zeros with minus one. Let the numbers of the small circle be $a_1, a_2, \dots, a_{140}$, and the numbers of the big circle be $b_1, b_2, \dots, b_{140}$. Note that in order for the corresponding pair to be a “good pair”, it is sufficient that the multiplication of these numbers equal to -1 . Then for Serdar, it is enough to find the number i to satisfy:

$$\sum_{k=1}^{140} a_k b_{i+k-1} \leq 0$$

Here $b_{j+140} = b_j$. This is because the sum of $a_k b_{i+k-1}$ is equal to 1 or -1 , and if this sum is less than or equal to 0, then the sum of at least 70 additions must be -1 .

According to this, at least 70 “good pairs” will be obtained if Serdar revolves around the numbers to form the corresponding pair. So, it is enough to prove that the whole number i can be found to make this sum negative. Let’s assume the opposite. That is, the sum obtained for any $i = 1, 2, \dots, 140$ be positive. Then, if we add these results the sum also should be positive. From here,

$$(a_1 + a_2 + \dots + a_{140})(b_1 + b_2 + \dots + b_{140}) > 0$$

Which is not possible. Because we have 140 ones, and 140 minus ones, and in one of the sequences $\{a_n\}_{n=1}^{140}$ and $\{b_n\}_{n=1}^{140}$ there is not less than 70 minus ones. Then, one of the sums of $\{a_n\}_{n=1}^{140}$ and $\{b_n\}_{n=1}^{140}$ is non-positive and other is non-negative, that is the multiplication of these sums cannot be positive.

Solution 2 (Jepbar Asgarov, Turkmen State Institute of Architecture and Construction):

Serdar will win in any situation. At the beginning, let there be x_1 number of “corresponding numbers”. Then if Serdar rotate inner circle by one step to clockwise let there be x_2 number of “corresponding number”. Then if Serdar go ahead by rotating the inner circle one step to clockwise order let the number of “corresponding numbers” after i^{th} rotation be $x_{(i+1)}$. It is trivial that if Serdar rotates inner circle by one step to clockwise order for 140 times, it will come to the initial position of inner circle. So the rotations after 139^{th} rotation are meaningless. Now let’s look to the sum of $x_1 + x_2 + \dots + x_{140}$ by another way.

Let there by k number of ones in inner circle, then there are $(140 - k)$ number of ones on outer circle. Likewise the number of zeros on inner circle is $(140 - k)$ and on outer is k .

Inner circle $\Rightarrow (140 - k)$ zeros and k ones.

Outer circle $\Rightarrow k$ zeros and $(140 - k)$ ones.

Now let's take some of the ones from inner circle. Let it be A . Then during the rotations of Serdar that A has corresponded exactly one time with each of the k zeros on outer circle. So the point gathered with A is equal to k . So, the point of A in $x_1 + x_2 + \dots x_{140}$ is k . But we have taken A randomly so each of the ones in inner circle has gathered k points. Overall all ones gathered $k * k$ points. Like above each zero which located on inner circle will gather $(140 - k)$ points and overall all zeros will gather $(140 - k) * (140 - k)$ points together.

So sum of all points $k * k + (140 - k) * (140 - k)$, but in other way it is equal to $x_1 + x_2 + \dots x_{140}$.

$$k * k + (140 - k) * (140 - k) = x_1 + x_2 + \dots x_{140}.$$

By Cauchy-Schwarz inequality:

$$(k^2 + (140 - k)^2)(1^2 + 1^2) \geq (k * 1 + (140 - k) * 1)^2 = 140^2$$

$$(k^2 + (140 - k)^2) \geq 2 * 70^2 = 70 * 140$$

So $x_1 + x_2 + \dots x_{140} \geq 70 * 140$. There are 140 number of summands on $L.H.S$. So, from Pigeonhole Principle, there is at least one i such that $x_i \geq 70$. So if Serdar rotates to $(i - 1)$ times to clockwise then he will reach a rotation with at least 70 number of “corresponding numbers”. So Serdar can guarantee winning.

Solution 3: (Akmyrat Tirkeshov, Institute of International Relations of the Ministry of Foreign Affairs of Turkmenistan)

Let 140 1s and 140 0s be placed on two circles. If the big circle has **a** 1s, then it has **140-a** 0s.

On the small circle, the number of 0s equal to **a**, and number of 1s equal to **140-a**.

Now I will keep the big circle in that position, and I will rotate the small circle 140 times so that each number on the big circle will be in line with the number on the small circle. Every 1s on my small circle form good pairs with every 0s on my big circle, just as every 0s on my small circle forms good pairs with every 1s on my big circle.

Then when I rotate the small circle 140 times, the total number of good pairs equal to:

$$(\mathbf{140 - a})^2 + \mathbf{a^2} = \mathbf{2 * (a - 70)^2 + 9800 \geq 9800}$$

Then I have at least 9800 good pairs when I rotate the small circle 140 times.

Since $9800:140=70$ then I have got at least once (in 140 times) more than 69 good pairs.

Serdar always have a winning strategy.

Team Olympiad

Problem 1 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous function such that

$$f(x) + f(-x) = \frac{x \sin x}{1 + \cos^2 x}$$

Then calculate integral:

$$\int_{-\pi}^{\pi} f(x) dx$$

Gurtmyrat Motukov (Magtymguly Turkmen State University, Turkmenistan)

Solution:

$$\begin{aligned} \int_{-\pi}^{\pi} f(x) dx &= \int_0^{\pi} (f(x) + f(-x)) dx = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx \\ &= -\frac{\pi}{2} (\arctan(\cos \pi) - \arctan(\cos 0)) = -\frac{\pi}{2} \left(-\frac{\pi}{4} - \frac{\pi}{4} \right) = \frac{\pi^2}{4} \end{aligned}$$

Hint: Here we have used well know equality from Calculus:

$$\int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(x) dx$$

Problem 2 Let $A(a_{ij}) \in M_n(\mathbb{R})$ matrix such that $a_{ij} = i^2 + j^2 + ij$. Calculate $\det(A)$

Pirmyrat Gurbanov (International University for the Humanities and Development, Turkmenistan)

Solution: If $n = 1, 2, 3$ then $\det(A) \neq 0$, so $\text{rank}(A) = n$

If $n \geq 4$, let $A = B + C + D$, where $b_{ij} = i^2$, $c_{ij} = j^2$ and $d_{ij} = ij$.

Then $\text{rang}(A) = \text{rang}(B + C + D) \leq \text{rang}(B) + \text{rang}(C) + \text{rang}(D) = 3 \Rightarrow \det(A) = 0$

Hint: Here we have used well know rank inequality:

$$\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$$

Problem 3 Let $z_1, z_2, \dots, z_n$ complex numbers ($n \geq 2$), the absolute values of which are equal r , but the sum is not equal to zero. Find the value of an expression:

$$\left| \frac{z_2 z_3 z_4 \dots z_n + z_1 z_3 z_4 \dots z_{n-1} z_n + \dots + z_1 z_2 z_3 \dots z_{n-1}}{z_1 + z_2 + \dots + z_n} \right|$$

Solution: Knowing the module, we represent complex numbers in exponential form $z_k = r e^{i\varphi_k}$, then:

$$\begin{aligned} \left| \frac{z_2 z_3 z_4 \dots z_n + z_1 z_3 \dots z_n + \dots + z_1 z_2 \dots z_{n-1}}{z_1 + z_2 + \dots + z_n} \right| &= \left| \frac{z_1 z_2 \dots z_n \frac{z_1^{-1} + z_2^{-1} + \dots + z_n^{-1}}{z_1 + z_2 + \dots + z_n}}{z_1 + z_2 + \dots + z_n} \right| \\ &= \left| r^n \frac{z_1^{-1} + z_2^{-1} + \dots + z_n^{-1}}{z_1 + z_2 + \dots + z_n} \right| = \left| r^n \frac{\sum_{k=1}^n r^{-1} e^{-i\varphi_k}}{\sum_{k=1}^n r e^{i\varphi_k}} \right| \\ &= r^{n-2} \left| \frac{\sum_{k=1}^n r e^{-i\varphi_k}}{\sum_{k=1}^n r e^{i\varphi_k}} \right| = r^{n-2}. \end{aligned}$$

Problem 4 Let n and i are integers such that $n \geq i \geq 0$ then prove that

$$0 < 1 - \frac{C_i}{2^{2i+1}} - \{10^{(2i+1)n} \sqrt{10^{2n} - 1}\} < \frac{1}{10^{2n}},$$

where $\{.\}$ denotes fractional part of the number and C_k are Catalan numbers: $C_0 = 1$ and

$$C_k = \frac{1}{k+1} \binom{2k}{k} \text{ for } k \geq 1.$$

Yagub Aliyev (ADA University, Azerbaijan)

Solution: Firstly we write Taylor's representation for the function,

$$f(x) = \sqrt{1-x}:$$

If $|x| < 1$ then we have $\sqrt{1-x} = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = 1 - \sum_{k=1}^{\infty} \frac{(2k-3)!!}{2^k k!} x^k = 1 - \sum_{k=1}^{\infty} \frac{2C_{k-1}}{4^k} x^k$. Thus we obtain,

$$10^{2n} - 10^n \sqrt{10^{2n} - 1} = 10^{2n} \left(1 - \sqrt{1 - \frac{1}{10^{2n}}} \right) = 10^{2n} \sum_{k=1}^{\infty} \frac{2C_{k-1}}{4^k 10^{2nk}} = \sum_{k=0}^{\infty} \frac{C_k}{4^k 10^{2nk} \cdot 2}.$$

Let $n > 0$ and designate the last sum as $a_n = \sum_{k=0}^{\infty} \frac{C_k}{4^k \cdot 10^{2nk} \cdot 2}$. Let us prove that integer part of the number $a_n \cdot 10^{2in}$ is $[a_n \cdot 10^{2in}] = \sum_{k=0}^{i-1} \frac{C_k 10^{2n(i-k)}}{4^k \cdot 2}$. For this we must prove that

- 1) $\sum_{k=0}^{i-1} \frac{C_k 10^{2n(i-k)}}{4^k \cdot 2}$ is an integer, and
- 2) $\sum_{k=i}^{\infty} \frac{C_k}{4^k \cdot 10^{2n(k-i)} \cdot 2} < 1$.

Proof of 1): We prove that each of the numbers $\frac{C_k 10^{2n(i-k)}}{4^k \cdot 2}$ in the sum $\sum_{k=0}^{i-1} \frac{C_k 10^{2n(i-k)}}{4^k \cdot 2}$ is an integer. Indeed, since $n \geq i > k$, $i - k \geq 1$ then $n \geq k + 1$ therefore, $2^{2k+1} | 2^{2k+2} | 2^{2n} | 2^{2n(i-k)} | 10^{2n(i-k)}$, where $a|b$ means a divides b . Catalan numbers C_k are integers, therefore all the numbers $\frac{C_k 10^{2n(i-k)}}{4^k \cdot 2}$ in the sum are integers.

Proof of 2): Note that $C_k \leq 4^k$. Then:

$$\sum_{k=i}^{\infty} \frac{C_k}{4^k \cdot 10^{2n(k-i)} \cdot 2} \leq \frac{1}{2} \sum_{k=i}^{\infty} \frac{1}{10^{2n(k-i)}} < \frac{1}{2} \cdot 2 = 1.$$

Now we study the fractional part of the number $a_n \cdot 10^{2in}$.

We proved that

$$\{a_n \cdot 10^{2in}\} = \sum_{k=i}^{\infty} \frac{C_k}{4^k \cdot 10^{2n(k-i)} \cdot 2} = \frac{C_i}{4^i \cdot 2} + \sum_{k=i+1}^{\infty} \frac{C_k}{4^k \cdot 10^{2n(k-i)} \cdot 2}.$$

The number $\frac{C_i}{4^i \cdot 2}$ is not dependent on n , and the remaining sum $\sum_{k=i+1}^{\infty} \frac{C_k}{4^k \cdot 10^{2n(k-i)} \cdot 2}$ can be evaluated noting that $n > 0$ and $\sum_{k=1}^{\infty} \frac{C_k}{4^k} = 1$.

$$0 < \sum_{k=i+1}^{\infty} \frac{C_k}{4^k \cdot 10^{2n(k-i)} \cdot 2} < \frac{1}{10^{2n} \cdot 2} \sum_{k=1}^{\infty} \frac{C_k}{4^k} = \frac{1}{10^{2n} \cdot 2} < \frac{1}{10^{2n}}.$$

We obtain that for $n \geq i > 0$ and $n > i = 0$, the following inequality is true:

$$\frac{C_i}{4^i \cdot 2} + \frac{1}{10^{2n}} > \{a_n \cdot 10^{2in}\} > \frac{C_i}{4^i \cdot 2}.$$

Nothing the definition of a_n we obtain

$$\frac{C_i}{4^i \cdot 2} + \frac{1}{10^{2n}} > \left\{ 10^{(2i+2)n} - 10^{(2i+1)n} \sqrt{10^{2n} - 1} \right\} > \frac{C_i}{4^i \cdot 2}.$$

Since the number $10^{(2i+2)n}$ is an integer then

$$\frac{C_i}{4^i \cdot 2} + \frac{1}{10^{2n}} > \left\{ -10^{(2i+1)n} \sqrt{10^{2n} - 1} \right\} > \frac{C_i}{4^i \cdot 2}.$$

Note that if t is not an integer then $\{-t\} = 1 - \{t\}$. Using this equality and noting that $\sqrt{10^{2n} - 1}$ is irrational for $n > 0$, we obtain

$$\frac{C_i}{4^i \cdot 2} + \frac{1}{10^{2n}} > 1 - \left\{ 10^{(2i+1)n} \sqrt{10^{2n} - 1} \right\} > \frac{C_i}{4^i \cdot 2}.$$

But surprisingly this inequality is true also for $n = i = 0$. Therefore the required inequality problem is true for all $n \geq i \geq 0$.

Problem 5 There are numbers $1, 2, \dots, n$ on the game roulette. How many times on average should you spin the reel to have the total amount of dropped numbers not less than n ?

Pirmyrat Gurbanov (International University for the Humanities and Development, Turkmenistan)

Solution: We will spin the reel until the sum of the dropped points become greater than or equal to n . Let a random variable ξ be the number of such rotations, $p_k = P(\xi = k)$. We need to find

$$M\xi = \sum_{k=1}^n kp_k.$$

Let's calculate the probability p_k in general cases. Let r_i points drop out at the i^{th} rotation of the reel ($i = 1, 2, \dots, k$). Then the number of all possible sets $(r_1, \dots, r_k)$ is exactly n^k , and the number of those for which

$r_1 + r_2 + \dots + r_k < n$ coincides with the number of natural solutions of the equation $x_1 + x_2 + \dots + x_k + x_{k+1} = n$. It is known that this equation has exactly C_{n-1}^k natural solutions. From here, $a_k = P(\xi > k) = C_{n-1}^k n^k$. Obviously, it's $P_n = a_{n-1}$, and for $k < n$ we have $P_k = P(\xi > k-1) - P(\xi > k) = a_{k-1} - a_k$.

Hence,

$$\begin{aligned} M\xi &= \sum_{k=1}^{n-1} k(a_{k-1} - a_k) + na_{n-1} = \sum_{k=1}^n a_{k-1} + \sum_{k=1}^n (k-1)a_{k-1} = \\ &= \sum_{k=1}^n a_{k-1} = \sum_{i=0}^{n-1} a_i = \sum_{i=0}^{n-1} C_{n-1}^i n^i \end{aligned}$$

According to Newton's binomial formula, this is equal to

$$\left(1 + \frac{1}{n}\right)^{n-1}$$

Problem 6 Let $A, B \in M_n(C)$ such that $A^2 = A$ and $B^2 = B$. Prove that:

$$Im(AB - BA) = Im(A + B - I) \cap Im(A - B)$$

where for $M \in M_n(C)$, we denote $ImM = \{MX : X \in M_{n,1}(C)\}$

Vasile Pop (Technical University of Cluj-Napoca, Romania)

Solution: Since $A^2 = A$ and $B^2 = B$, it can be easily checked that

$$(A - B)(A + B - I_n) = A \cdot B - B \cdot A \quad (1)$$

$$(A + B - I_n)(A - B) = -(A \cdot B - B \cdot A). \quad (2)$$

$$(2A - I_n)(A \cdot B - B \cdot A) = -(A \cdot B - B \cdot A)(2A - I_n) = AB + BA - 2ABA. \quad (3)$$

By (1) and (2),

$$Im(A \cdot B - B \cdot A) \subset Im(A - B)$$

$$Im(A \cdot B - B \cdot A) \subset Im(A + B - I_n),$$

Hence,

$$Im(A \cdot B - B \cdot A) \subset Im(A + B - I_n) \cap Im(A - B).$$

Conversely, let $X \in Im(A + B - I_n) \cap Im(A - B)$. Then

$$X = (A + B - I_n)Y \quad \text{and} \quad X = (A - B)Z, \quad Y, Z \in M_{n,1}(C).$$

It follows that

$$(A - B)X = (A - B)(A + B - I_n)Y = (A \cdot B - B \cdot A)Y$$

And

$$(A + B - I_n)X = (A + B - I_n)(A - B)Z = -(A \cdot B - B \cdot A)Z.$$

Adding the last two relations, we obtain

$$(2A - I_n)X = (A \cdot B - B \cdot A)(Y - Z), \quad (4)$$

Which leads to

$$(2A - I_n)^2 \cdot X = (2A - I_n)(A \cdot B - B \cdot A)(Y - Z). \quad (5)$$

We conclude by (3), (4) and (5) that

$$X = (A \cdot B - B \cdot A) \cdot (2A - I_n)(Z - Y) \in Im(A \cdot B - B \cdot A).$$

The proof is now concluded.

Problem 7 Determine all injective functions $f: [2020, 2021] \rightarrow [2020, 2021]$ satisfying

$$f(-3f(x) + f(f(x)) + 3x) = x \quad \text{for all } x \in [2020, 2021].$$

Note: $f(x)$ function is called injective if $f(x_1) = f(x_2)$ then $x_1 = x_2$.

Palvan Agamyradov (Magtymguly Turkmen State University, Turkmenistan)

Solution 1: If we put $x = f(x)$ given equation then we get

$$f(-3f(f(x)) + f(f(f(x))) + 3f(x)) = f(x)$$

from the given function is injective

$$-3f(f(x)) + f(f(f(x))) + 3f(x) = x,$$

$$f(f(f(x))) - 3f(f(x)) + 3f(x) - x = 0 \quad (*)$$

We take some $x_0 \in [2020, 2021]$ and $x_n = f(x_{n-1})$ for $n = 1, 2, \dots$. Setting $x = x_n$ in $(*)$ then we obtain

$$x_{n+3} - 3x_{n+2} + 3x_{n+1} - x_n = 0$$

From the last equation we can write

$$x_{n+3} - 2x_{n+2} + x_{n+1} = x_{n+2} - 2x_{n+1} + x_n.$$

This equation is invariant and we can get

$$x_{n+3} - 2x_{n+2} + x_{n+1} = x_2 - 2x_1 + x_0.$$

If we put $n = 0, 1, \dots, k$ to the last equation and add them

$$x_{k+3} - x_{k+2} - x_1 + x_0 = (k+1)(x_2 - 2x_1 + x_0).$$

Since $|x_{k+3} - x_{k+2} - x_1 + x_0| \leq 2$ ($2020 \leq x_i \leq 2021$) we conclude that $|x_2 - 2x_1 + x_0| \leq \frac{2}{k+1}$ for all $k \geq 0$. Therefore $x_2 - 2x_1 + x_0 = 0$ that is $f(f(x_0)) - 2f(x_0) + x_0 = 0$ for all $x_0 \in [2020, 2021]$. We can write this like

$$f(f(x)) - 2f(x) + x = 0 \text{ for all } x \in [2020, 2021].$$

Here we do the same thing, that is we fix $y_0 \in [2020, 2021]$ and $y_n = f(y_{n-1})$ for $n = 1, 2, \dots$. Then we get

$$y_{n+2} - 2y_{n+1} + y_n = 0 \text{ and } y_{n+2} - y_{n+1} = y_{n+1} - y_n = \dots = y_1 - y_0$$

and from here $y_n - y_0 = n(y_1 - y_0)$. Since $|y_n - y_0| \leq 1$, we conclude that $|y_1 - y_0| \leq \frac{1}{n}$ for all $n \geq 1$ that is $y_1 = y_0$. This argument shows that the only solution is given by $f(x) = x$, for all $x \in [2020, 2021]$. This satisfies the given equation.

Solution 2: Similar with first solution to the following equation:

$$x_{n+3} - 3x_{n+2} + 3x_{n+1} - x_n = 0.$$

Characteristic equation of this recurrence relation is $x^3 - 3x^2 + 3x - 1 = 0$ and all solution of this equation $x = 1$. From here we get $x_n = a + bn + cn^2$ for some $a, b, c \in R$. From the range of the given function all elements of the sequence $\{x_n\}$ must be lie in the interval $[2020, 2021]$. It follows that $b = c = 0$ and the sequence is constant: $x_n = a$. As a result, $x_1 = x_0$ and $f(x_0) = x_0$. This gives us only one solution $f(x) = x$.

Problem 8 Let G be group and f mapping is its second order automorphism. If unit element of group G is its unique fixed element and for every $x \in G$ there is a single element $y \in G$ such that $x = y^2$ then prove that G is Abel group.

Pirmyrat Gurbanov (International University for the Humanities and Development, Turkmenistan)

Solution: Consider the mapping $g(x) = x^{-1}f(x)$ for every $x \in G$. Let us to show that g is injective: $g(x) = g(y) \Rightarrow x^{-1}f(x) = y^{-1}f(y) \Rightarrow f(xy^{-1}) = xy^{-1}$

$$\Rightarrow xy^{-1} = e \Rightarrow x = y$$

Let $x \in G$. Then exist unique element $y \in G$ with property $x^{-1}f(x) = y^2$. We deduce that $f(x^{-1}f(x)) = f(y^2) \Leftrightarrow f(x)^{-1}f(f(x)) = f(y)^2 \Leftrightarrow f(x)^{-1}x = f(y)^2$

$$\Leftrightarrow (f(x)^{-1}x)^{-1} = (f(y)^2)^{-1} \Leftrightarrow x^{-1}f(x) = f(y^{-1})^2$$

From hypothesis we have $f(y^{-1}) = y \Leftrightarrow f(y) = y^{-1}$.

Calculating $f(xy) = f(x)f(y) = xy^2y^{-1} = xy \Rightarrow xy = e \Leftrightarrow y = x^{-1} \Rightarrow x^{-1}f(x) = y^2 = x^{-2} \Rightarrow f(x) = x^{-1}$ for every $x \in G$.

Then from here we have for every $x, y \in G$

$$xy = f(f(xy)) = f(f(x)f(y)) = f(x^{-1}y^{-1}) = f((yx)^{-1}) = yx$$

Problem 9 Calculate limit

$$\lim_{n \rightarrow \infty} e^{-n} \left(\sum_{k=0}^n \frac{n^k}{k!} \right)$$

Gurtmyrat Motukov (Magtymguly Turkmen State University, Turkmenistan)

Solution: Consider the function $f: [0, \infty) \rightarrow \mathbb{R}$ given by $f(x) = xe^{-x}$.

Let's first prove that, for every $\varepsilon \in (0,1)$, there exists $\delta \in (0,1)$ such that,

$\forall x \in (1 - \delta, 1 + \delta)$, one has:

$$e^{-1 - \frac{(x-1)^2}{2}(1+\varepsilon)} \leq f(x) \leq e^{-1 - \frac{(x-1)^2}{2}(1-\varepsilon)}. \quad (1)$$

Observe that f achieves a strict maximum at 1 and that $f(0) = 0$ and $\lim_{x \rightarrow \infty} f(x) = 0$.

Consider $g: (0, \infty) \rightarrow \mathbb{R}$ given as $g(x) = \ln(f(x))$. Remark that $g(1) = -1$, $g'(1) = 0$ and $g''(1) = -1$. Then, by the Taylor formula, there exists a function α such that $\lim_{x \rightarrow 1} \alpha(x) = 0$ such that

$$g(x) = -1 - \frac{(x-1)^2}{2} (1 + \alpha(x)).$$

Using $\lim_{x \rightarrow 1} \alpha(x) = 0$ it follows that for every $\varepsilon \in (0, 1)$, there exists $\delta \in (0, 1)$ such that, for every $x \in (1 - \delta, 1 + \delta)$, one has $|\alpha(x)| < \varepsilon$. Now, let's denote

$$\iota = \lim_{n \rightarrow \infty} \frac{1}{n!} \cdot \int_0^n f(t) \cdot t^{n-1} dt. \quad (2)$$

By the change of the variable $nt \rightarrow t$ one has

$$\iota = \lim_{n \rightarrow \infty} \frac{n^{n+1}}{n!} \cdot \int_0^1 (xe^{-x})^n dx.$$

Denote

$$I_n = \int_0^1 (f(x))^n dx = \int_0^1 (xe^{-x})^n dx.$$

Observe that in view of 1), that for every $\varepsilon \in (0, 1)$, there exists $\delta \in (0, 1)$ such that

$$I_n \geq \int_{1-\delta}^1 (xe^{-x})^n dx \geq \int_{1-\delta}^1 e^{-n - \frac{(x-1)^2}{2}(1+\varepsilon)n} dx,$$

Hence, using also the Stirling formula, there exists $\theta_n \rightarrow 1$ such that

$$\frac{n^{n+1}}{n!} I_n = \frac{n^{n+1}}{\sqrt{2n\pi} \left(\frac{n}{e}\right)^n \theta_n} I_n \geq \frac{n^{n+1}}{\sqrt{2n\pi} \left(\frac{n}{e}\right)^n \theta_n} \cdot \frac{1}{e^n} \cdot \int_{1-\delta}^1 e^{-\frac{(x-1)^2}{2}(1+\varepsilon)n} dx.$$

Now, observe that, by the change of the variable $\sqrt{(1+\varepsilon)n}(x-1) = -y$, the integral $\int_{1-\delta}^1 e^{-\frac{(x-1)^2}{2}(1+\varepsilon)n} dx$ becomes $\frac{1}{\sqrt{(1+\varepsilon)n}} \int_0^{\delta\sqrt{(1+\varepsilon)n}} e^{-\frac{y^2}{2}} dy$.

It follows that

$$\frac{n^{n+1}}{n!} I_n \geq \frac{1}{\theta_n \sqrt{2\pi(1+\varepsilon)}} \int_0^{\delta\sqrt{(1+\varepsilon)n}} e^{-\frac{y^2}{2}} dy.$$

Passing to the limit for $n \rightarrow \infty$ and using the fact that $\int_0^\infty e^{-\frac{y^2}{2}} dy = \frac{\sqrt{2\pi}}{2}$, we obtain that

$$\iota \geq \frac{1}{2\sqrt{1+\varepsilon}}.$$

For the upper bound for I_n , observe that

$$I_n - \int_{1-\delta}^1 (xe^{-x})^n dx = \int_0^{1-\delta} (xe^{-x})^n dx \leq \left(\max_{x \in [0, 1-\delta]} f(x) \right)^{n-1} \cdot I_1,$$

Since $m_\delta := \max_{x \in [0, 1-\delta]} f(x) < f(1) = \frac{1}{e}$, it follows that

$$I_n \leq \int_{1-\delta}^1 (xe^{-x})^n dx + m_\delta^{n-1} \cdot I_1 = (m_\delta e)^n \cdot \frac{1}{e^n} \cdot \frac{I_1}{m_\delta},$$

Hence for every $\varepsilon \in (0, 1)$, there exists $\delta \in (0, 1)$ such that, $A := m_\delta e \varepsilon \in (0, 1)$ and $k := \frac{I_1}{m_\delta} > 0$, such that

$$I_n \leq \int_{1-\delta}^1 (xe^{-x})^n dx + k \cdot \left(\frac{A}{e} \right)^n.$$

Then, reasoning as above, it follows that

$$\begin{aligned} \frac{n^{n+1}}{n!} I_n &= \frac{n^{n+1}}{\sqrt{2n\pi} \left(\frac{n}{e} \right)^n \theta_n} I_n \leq \frac{n^{n+1}}{\sqrt{2n\pi} \left(\frac{n}{e} \right)^n \theta_n} \cdot \left[\frac{1}{e^n} \int_{1-\delta}^1 e^{-\frac{(x-1)^2}{2}(1-\varepsilon)^n} dx + k \cdot \left(\frac{A}{e} \right)^n \right] = \\ &= \frac{1}{\theta \sqrt{2\pi(1-\varepsilon)}} \int_0^{\delta \sqrt{(1-\varepsilon)n}} e^{-\frac{y^2}{2}} dy + \frac{k}{\sqrt{2\pi\theta_n}} \cdot \sqrt{n} \cdot A^n. \end{aligned}$$

Passing to the limit for $n \rightarrow \infty$, it follows that $\iota \leq \frac{1}{2\sqrt{1-\varepsilon}}$.

In conclusion,

$$\frac{1}{2\sqrt{1+\varepsilon}} \leq \iota \leq \frac{1}{2\sqrt{1-\varepsilon}},$$

For every $\varepsilon \in (0, 1)$, hence $\iota = \frac{1}{2}$.

Now observe, integrating by parts, that

$$\int_0^n e^{-t} t^n dt = n! - e^{-n} (n^n + n \cdot n^{n-1} + n(n-1) \cdot n^{n-2} + \dots + n!).$$

Using 2), it follows that

$$\frac{1}{2} = \lim_{n \rightarrow \infty} \frac{1}{n!} \cdot \int_0^n e^{-t} t^n dt = 1 - \lim_{n \rightarrow \infty} e^{-n} \cdot \left(\sum_{k=0}^n \frac{n^k}{k!} \right).$$

Hence, the desired limit equals to $\frac{1}{2}$.

Problem 10 Let α and β be the positive numbers satisfying the following equations:

$$\alpha^2 - 2019\alpha - \frac{2019}{2020} = 0 \quad \text{and} \quad \beta^2 - 2019\beta - \frac{2020}{2019} = 0,$$

Prove the next inequality

$$\lim_{n \rightarrow \infty} \inf(n^2 \|n\alpha\| \|n\beta\|) < 2019.5^2$$

where $\|a\|$ is the distance between a and the nearest integer to a , for example

$$\|3.2\| = \|9.8\| = 0.2.$$

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Solution: We can easily see the following equalities:

$$\alpha^2 - 2019\alpha - \frac{2019}{2020} = 0 \Leftrightarrow (2020\alpha + 1)(\alpha - 2019) = \alpha \Leftrightarrow \alpha = 2019 + \frac{\alpha}{2020\alpha + 1}$$

Consequently,

$$\begin{aligned} \frac{1}{\alpha} &= \frac{1}{2019 + \frac{1}{2020 + \frac{1}{\alpha}}} = \frac{1}{2019 + \frac{1}{2020 + \frac{1}{2019 + \frac{1}{\alpha}}}} = \dots = \\ &= \frac{1}{2019 + \frac{1}{2020 + \frac{1}{2019 + \dots}}} =: [2019, 2020, 2019, 2020 \dots]. \end{aligned}$$

Similarly,

$$\frac{1}{\beta} = [2020, 2019, 2020, 2019 \dots].$$

It is not difficult to see that

$$\frac{1}{\beta} = 2019 + \alpha \tag{1}$$

Note that $\|\gamma + k\| = \|\gamma\|$, where $k \in \mathbb{Z}$. So by using (1) we have

$$\|n\alpha\| = \|n\alpha + 2019n\| = \left\| \frac{n}{\beta} \right\|$$

and

$$\|n\beta\| = \left\| \frac{n\beta^2}{\beta} \right\| = \left\| n \frac{2020\beta + \frac{2020}{2019}}{\beta} \right\| = \left\| \frac{2020n}{2019\beta} \right\|$$

So

$$n^2 \|n\alpha\| \|n\beta\| = n^2 \left\| \frac{n}{\beta} \right\| \left\| \frac{2020n}{2019\beta} \right\| = n^2 \|n\theta\| \left\| \frac{2020}{2019} n\theta \right\|, \quad (2)$$

Where $\theta = \frac{1}{\beta}$.

Lemma. Let $[a_1, a_2, \dots, a_n] = \frac{P_k}{Q_k}$ is a sequence of continued fraction, where $a_j \in \mathbb{N}$.

Then

$$\frac{P_{k+1}}{Q_{k+1}} = \frac{a_{k+1}P_k + P_{k-1}}{a_{k+1}Q_k + Q_{k-1}}.$$

Proof. We know that $\theta = \frac{1}{\beta} = [2020, 2019, 2020, 2019, \dots]$. Let

$$c_k := \frac{p_k}{q_k} = \left[2020, 2019, \dots, 2020 - \frac{1 + (-1)^k}{2} \right].$$

By the above Lemma we have the following equality

$$c_{k+1} = \frac{a_{k+1}p_k + p_{k-1}}{a_{k+1}q_k + q_{k-1}},$$

Where $a_{k+1} = 2020 - \frac{1+(-1)^k}{2}$. It is easy to see that $\{c_{2j-1}\}_{j \geq 1}$ decreases and $\{c_{2j}\}_{j \geq 1}$ increases. Since $\lim_{k \rightarrow \infty} c_k = \theta$, we have

$$c_{2j} < \theta < c_{2j+1}, \quad \forall j \geq 1.$$

Consequently,

$$|\theta - c_k| < |c_{k+1} - c_k| = \frac{1}{q_k q_{k+1}}.$$

Since $q_{k+1} = a_{k+1}q_k + q_{k-1}$, we have

$$q_{k+1} < 2021q_k.$$

Consequently,

$$\left(2019 + \frac{1}{2021} \right) q_k < q_{k+1}$$

For $k > 2$. Hence

$$q_k |q_k \theta - p_k| < \frac{q_k}{q_{k+1}} < \frac{1}{2019 + \frac{1}{2021}} \quad (3)$$

By using (2) and (3) and some simple inequalities we get

$$\begin{aligned} \lim_{n \rightarrow \infty} \inf n^2 \|n\alpha\| \|n\beta\| &= \lim_{n \rightarrow \infty} \inf n^2 \|n\theta\| \left\| \frac{2020}{2019} n\theta \right\| \\ &\leq \lim_{m \rightarrow \infty} \inf (2019m)^2 \|2019m\theta\| \|2020m\theta\| \\ &\leq \lim_{k \rightarrow \infty} \inf (2019q_k)^2 \|2019q_k\theta\| \|2020q_k\theta\| \\ &\leq \lim_{k \rightarrow \infty} \inf 2019^3 2020 q_k^2 \|q_k\theta\|^2 \\ &= \lim_{k \rightarrow \infty} \inf 2019^3 2020 q_k^2 |q_k\theta - p_k|^2 \\ &\leq \frac{2019^3 2020}{\left(2019 + \frac{1}{2021}\right)^2} < 2019.5^2. \end{aligned}$$